Research Statement

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I am an applied microeconomist working in development, labour and public economics. Who gets ahead in Latin America still depends heavily on family background, place of birth and ethnic identity, and I study how to measure these inequalities when data are incomplete and how they respond to policy. The standard tools often do not quite fit these questions, so I adapt them or build new ones, and that is where my comparative advantage lies. My thesis develops applied microeconometric methods for distributional analysis when data are scarce. Beyond the thesis, I apply them, together with causal designs, to inequality of opportunity, social mobility and the gaps Indigenous people face in Mexico.

Methods are what I bring to most projects. My master's in data science added machine learning to my training in econometrics. I code in R and Python and can also work in Stata, and I release my methods as open-source R packages. With Louis Sirugue (LSE), I co-maintain the GEOM database of inequality of opportunity estimates, and when a question needs data that do not exist yet, I collect them from text and web sources.

Outside academia, I work closely with policy institutions. Two 2025 reports build on work I have contributed to, CEEY's report on social mobility in Mexico and the OECD's report on inequality of opportunity. With CEEY, I also write for general audiences, including essays in Este País on inequality of opportunity and educational mobility.

1. PhD thesis: Essays on data scarcity and distributional analysis

Poverty, inequality and mobility statistics all depend on the whole shape of the welfare distribution, and social and public policy rely on them. High-income countries have administrative registers and large linked datasets that make these distributions observable. Latin America and most of the world lack that infrastructure, as do many regions within richer countries, so researchers rely on household surveys, which run infrequently and have small samples and limited geographic coverage. I refer to this as data scarcity, where data exist but not in the form or at the scale a distributional question needs.

The thesis takes on three versions of this problem. A single dataset may never record two variables together, the sample may be too small for the level of detail a question needs, or welfare data may exist only at a coarser geographic level than policy requires. In each case I develop a method that gets more out of the available sources and lets the data decide the shape of the distribution. I am the sole author of all three papers.

Job market paper

Bias Reduction in TSTSLS Applications: A Quantile Adjustment Method · R package tstsqa

The job market paper began while I was helping CEEY produce its 2025 report on social mobility in Mexico. Its main survey, ESRU-EMOVI, does not record income, so Luis Monroy-Gómez-Franco (University of Massachusetts Amherst), Roberto Vélez-Grajales (CEEY) and I imputed it from Mexico's national income and expenditure survey ("Survey-to-Survey Imputation When External Covariates Matter: Estimating Inequality of Opportunity in Mexico", CEEY working paper, 2025), and the report uses that imputation. This two-sample two-stage (TSTS) imputation predicts income from characteristics both surveys record, and I show that it distorts estimates in two ways. The predictions vary less than actual income, and they miss any link with the second variable that those characteristics do not capture. Existing corrections, such as stochastic augmentation, fix these problems one at a time, and fixing only the first can make an estimate worse.

I propose a quantile adjustment that replaces each prediction with the observed income at the same rank. This restores the income distribution and, in the same step, shifts the covariance with the second variable. I characterise when this shift reduces bias and provide two diagnostics that tell researchers in advance whether the adjustment will help. In applications to Mexico, Ecuador and Chile, it comes closest to the full-sample benchmark on 14 of 19 estimands, and researchers can apply it with my R package tstsqa. With Selçuk Bedük (University of Oxford) and H. Xavier Jara (LSE), I am now using it to impute household expenditure into the UK Household Longitudinal Study. I plan to submit the paper to the *Review of Economics and Statistics*.

Intergenerational mobility

The second paper began with a question about how intergenerational mobility differs across ethnic identities in Mexico, and where poverty traps appear. Answering it needs transition matrices on a fine grid. Quintile matrices are reliable but too coarse for the question, while percentile matrices leave most cells in a typical sample nearly empty. Headcount and kernel-based estimates then become unreliable in the tails, and parametric models impose functional forms that can distort the patterns.

I develop the positional functions framework in "Estimating Intergenerational Mobility: A Machine Learning Approach". For each person, it estimates a probability distribution over income destinations, given parental position and observed characteristics. Adding up these distributions gives non-parametric transition matrices on any grid, with covariates, that decompose exactly into subgroup components. With Swedish registers covering 1.3 million father-offspring pairs, I show that the framework matches established methods when data are abundant and, in smaller samples, estimates fine-grained matrices with less bias in the tails than headcount and kernel-based estimates. My target journal is the *Journal of Business & Economic Statistics*.

Poverty and inequality maps

Policymakers often need poverty and inequality statistics for municipalities, where household surveys have too few observations. Poverty maps solve this with small area estimation, fitting a welfare model in the survey and predicting welfare for every household in the census. Standard implementations such as ELL, however, assume the same parametric distribution in every area, which leads them to understate inequality.

Transformation forests relax that assumption. They estimate the welfare distribution locally, so both its shape and the effects of covariates can vary across areas, and a single model gives poverty rates, Gini coefficients and other indicators for each municipality. I am benchmarking the approach against the ELL and GAMLSS estimators with a simulation study and Mexico's 2025 intercensal survey, and I will submit the paper to the *Journal of Development Economics* once the benchmark is complete.

Ongoing methods work

I am also developing exact rank-level decompositions of changes in the Gini coefficient. The paper builds on Ferreira, Firpo and Galvao (2019).

2. Research beyond the thesis

Inequality of opportunity and mobility

Inequality of opportunity is the part of inequality that comes from circumstances people do not choose, such as their parents' education or where they grew up. I study it because it is the hardest part of inequality to justify, and because it ties inequality today to mobility across generations. That link is the focus of the programme on Opportunity, Mobility and the Intergenerational Transmission of Inequality at LSE's International Inequalities Institute, where I do most of this work.

My paper with Thibaut Plassot (Bordeaux School of Economics) and Isidro Soloaga (Universidad Iberoamericana), "Inequality of Opportunity in Mexico and its Regions: A Data-Driven Approach" (*The Journal of Development Studies*, 2022), measures inequality of opportunity in Mexico with conditional inference trees. I designed the estimation and wrote the first draft. Parental education and wealth, and the area where people grew up, are the main circumstances, and inequality of opportunity is highest in rural areas and in the South and Centre.

I am one of 18 co-authors of "International Variation in Inequality of Opportunity: Insights from 72 Countries", under revise and resubmit at *Nature*, a project that Paolo Brunori (LSE), Francisco Ferreira (LSE), Vito Peragine (University of Bari) and Pedro Salas-Rojo (CUNEF) lead. The paper estimates inequality of opportunity for 72 countries from 196 household surveys, and the GEOM database makes the estimates public. I built the R functions behind the estimates with Brunori and Salas-Rojo and cleaned part of the survey data. The OECD's 2025 report on inequality of opportunity, *To Have and Have Not*, builds on the GEOM methodology.

Mobility was also the subject of my master's thesis. I used a sharp regression discontinuity design around Mexico's 1993 extension of compulsory schooling and found no immediate effect on intergenerational mobility, although the trends pointed to a possible longer-term one.

A current project extends the measurement of inequality of opportunity to a multidimensional setting. I use distribution regression forests for multiple outcomes to find types from the joint distribution of income, schooling and health in Chile, and then decompose inequality of opportunity into a part between types and a part within them.

Indigenous identity

Indigenous people in Mexico still face large gaps in education and income, and understanding why has interested me since my undergraduate thesis on Indigenous self-identification and wage gaps.

That thesis became my paper with Eva Arceo-Gómez (CIDE), who supervised it ("Brechas salariales por autoidentificación indígena y rasgos lingüísticos en México", *Sobre México*, 2021). We estimate a switching regression and find that most of the average wage gap comes from selection into self-identification, while an unexplained gap remains that may reflect discrimination.

Guillermo Woo-Mora (Paris School of Economics) invited me to revisit Bonfil Batalla's *México Profundo* (1987) with him in "New Echoes of Deep Mexico? Indigenous Identity and Its Enduring Imprints" (chapter in *Roots of Underdevelopment*, Volume 2, 2026). With historical census tabulations and harmonised microdata, we show that education and income gaps have persisted over time and examine skin tone, historical Indigenous presence and Indigenous institutions as channels through which identity still affects economic outcomes.

An ongoing project uses census data to study whether labour market discrimination against Indigenous workers affects the decision to speak an Indigenous language and to pass it on to children. The framework from my second paper will also let me compare mobility across ethnic identities at the level of detail that first motivated it.

Other collaborations

Colleagues ask me to join projects outside these areas to design the identification strategy and the estimation. Sandra Aguilar-Gómez (Tecnológico de Monterrey), Eva Arceo-Gómez, Elia De la Cruz Toledo (Loyola Marymount University) and I study compliance with stay-at-home orders in Mexico during COVID-19 ("Follow the Leader? Public Health Messaging and Containment of Mobility During a Pandemic", UNDP working paper, 2022). I collected the data from the daily government briefings and from web sources, modelled their topics with natural language processing, and designed and estimated the quantile regressions that link the messages to municipal mobility.

With Heidi Smith (Universidad Iberoamericana) and Isabel Melguizo (CUNEF), I am studying whether political polarization affects how much Mexican municipalities borrow. I designed the identification strategy, built the panel from public finance, electoral and party ideology data, and ran the estimations.

Next steps

I plan to keep developing flexible models of the conditional income distribution and rank-based methods for distributional analysis. I want to use them to study how inequality and poverty persist across generations and groups in Latin America, with a focus on the Indigenous population in Mexico.